PEER REVIEW

MScFE Capstone Project

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Objectives

The authors describe an approach to measure the effects of market shocks such as the COVID-19 pandemic and Russian-Ukraine conflict on commodities, forex exchanges, equity markets, and uncertainty in the US and Australian financial markets. The article focuses on price data from these financial markets to construct a dynamic topology of market interconnectedness similar to a recently published paper given in this reference by Chengying et al. Chengying et al describe studying network topology constructed by price data to draw how interconnected the markets are and how market shocks like Covid 19 contribute to systemic risk, which can be identified using a similar approach as the authors in this article.

Authors apply the Quantile Time-Varying Vector Autoregression (TVP-VAR) method to understand the relationship among commodity prices such as gold and oil, forex rates, and equity markets. The TVP-VAR method became very popular when this model was developed by Primiceri (2005) showed application where he reliably predicted the time-varying structure of monetary policy transmission effects on the economy. The authors' approach using TVP-Var is to construct the network topology and identify how shocks in one market (for example in commodities) can propagate to others for example forex markets etc.

Theory:

The TVP-VAR (Quantile Time-Varying Parameter Vector Autoregression) approach used in the study has several assumptions, weaknesses, and potential gaps:

Assumptions:

Time-Varying Relationships: The TVP-VAR model assumes that the relationships between financial markets evolve over time. This allows for dynamic interdependencies, which can capture changes during normal and extreme market conditions.

Quantile-Based Analysis: The model assumes that financial market variables exhibit different behaviors at different quantiles (e.g., during extreme downturns or booms). It’s expected to perform well in both average and extreme scenarios, making it a robust tool for tail risk analysis.

Weaknesses or Gaps:

Complexity and Computational Burden: The TVP-VAR model is complex and computationally intensive, particularly when applied to high-dimensional datasets, such as a large number of financial assets or indicators over long periods. This can make estimation and interpretation more difficult, especially for real-time analysis.

Limited Structural Insights: While the TVP-VAR model can capture time-varying relationships, it doesn’t necessarily provide a clear structural interpretation of why or how these relationships evolve. It focuses on correlations and co-movements without delving into the underlying mechanisms driving these changes.

Sensitivity to Model Specifications: The performance and results of the TVP-VAR model are highly sensitive to the choice of parameters, including the number of lags, quantiles, and the way shocks are modeled. Incorrect specifications can lead to misleading results, particularly when analyzing extreme events.

Potential Gaps:

Neglect of Nonlinearities: Financial markets often exhibit nonlinear behaviors, especially during crises or extreme events. While TVP-VAR captures dynamic relationships, it may not fully account for nonlinear dependencies between variables.

Limited Focus on External Factors: The study emphasizes price data but might not incorporate other relevant external factors like policy decisions, geopolitical tensions, or technological changes that could significantly influence market dynamics beyond what is captured in historical price movements.

Concepts

Strengths of the TVP-VAR Model

Capturing market Shocks

TVP models have been demonstrated to capture market shocks by modeling the evolving price volatilities etc.

Topological Analysis of Markets.

TVP-VAR models allow the construction of network topology. This enables us to visualize how financial markets are interconnected. For example within a financial market like commodities how each type of commodity is connected to the other. In the given reference by Chengying, et al, they have constructed network topologies using a minimum spanning trees approach. Then this connected network topology can be used to assess how the systemic risk spreads during periods of significant market volatility and market shock events like the Covid-19 period.

Weaknesses of the TVP-VAR model:

Complexity and Interpretability:

After the construction of the network topology, it can be difficult to interpret the results on why those patterns exist among a set of variables and the exact reasons for the time-varying relationship between variables.

Sensitivity to Parameter Choices:

TVP-VAR model typically uses inputs such as the number of lags, quantiles, and correlation parameters related to the time-varying nature of independent variables such as commodities, forex, etc. If these inputs are poorly chosen, the constructed topological model can show misleading results and overfitting.

Correlation vs. Causation:

TVP-VAR models are based on correlations between price movements and volatility of independent variables. Studying these types of movements does not provide insights into causal relationships between these variables. For example, the model may predict a certain commodity, and forex may move together or have a high correlation of close to 1. But does not explain the reasons behind such a move. This limits the ability to identify the true drivers of the market changes during market shock studies and therefore can become less informative when developing economic strategies.

Contribution or extension of the topic

The authors focus on extreme quantiles of market behavior to gain insights into commodities, forex, and other market conditions to gain insights instead of focusing on just average conditions. This is a key advantage over other mean-variance optimization or techniques that focus on mean-based volatility of price effect methods.

This type of study on extreme conditions is important in quantifying the tail risks in market portfolio returns during crisis periods such as in COVID-19 and the Ukraine-Russia War.

Also, the authors included commodities such as oil and gold. Gold is typically considered a hedging instrument during market shock events. Oil price changes convey consumer sentiment in the economy. Therefore the model and results can be used as early warning indicators of market instability and to identify early signs of systemic risk.

Methodology (strengths or gaps)

Strengths

The methodology advocates the use of TVP-VAR, traditional mean-based analysis ignores the extremes - TVP-VARs ability to capture the dynamics of extremes makes it highly appealing to explore various market relationships (relations between commodities, forex exchanges, equity markets, and uncertainty) at the extremes.

The methodology addresses the dynamic nature of markets - markets are essentially dynamic, shocks propagate through the system, and asset classes influence each other in a correlated way affecting volatility across the markets.

The methodology incorporates behavioral finance - investor sentiment and psychology play an important role in determining asset prices. By including sentiment indicators, the methodology seeks to acknowledge a key behavioral market indicator in determining oil prices.

The methodology relies on quantitative metrics for establishing connectedness - both static and dynamic. Relies on metrics such as the total Connected Index (TCI), Directional Connectedness, and Dynamic Connectedness Measures.

Gaps

The TVP-VAR is presented as a tool to capture the dynamic interdependence but fails to show which specific quantiles will be examined under extreme market conditions or even how the time-varying parameters will be determined.

The article does not elaborate on how the outputs will be generated and interpreted and for example how policy makers might use this.

Regarding data transformation, the article mentions pre-processing data using a min-max scaler for normalizing but doesn't elaborate on dealing with outlier data or how cointegrated data series impact the results and how this should be mitigated.

Literature review (weaknesses or gaps)

The authors mention conventional methodologies of risk assessment, such as correlation analysis or traditional VaR, and mention that these don't cover interdependence among various market factors such as commodities, forex, equity, and uncertainty. However, the paper does not provide specific examples of these shortcomings.

The authors mention that they intend to cover US and Australian markets and to provide regional insights, however, they do not cite many articles that cover Australian markets.

Also, the paper focuses much more exclusively on studies related to TVP-VAR or similar methodologies such as quantile connectedness, the paper similarly does not explore literature related to market sentiment and oil prices.

Style

The authors use a single-column paper template for the project report. It begins with an abstract, followed by seven sections: introduction, theoretical framework, methodology, data, expected results, conclusion, and source codes. The report also includes references at the end.

The authors have presented a style of scientific journal article with formal academic English throughout the report. Two important components in a scientific paper are the theoretical framework and methodology sections; the former demonstrates the literature work from different authors, and the latter explains the mathematical formulas and the logic behind the Time-Varying Parameter Vector Autoregression model. Data was presented with a table showing the type, frequency, and source. Overall, the report adopted a typical scientific paper format with financial academics as its target audience.

Results and conclusions (weaknesses or gaps)

The authors provided a brief description of the expected results, probably because the model is still in the early stages. Although we understand that this report aims to uncover insights during extreme market conditions, a detailed explanation regarding how this can be reflected in expected results might be more helpful.

The conclusion section primarily focused on emphasizing the project's strengths such as improving the knowledge of extreme interdependencies across financial markets and helping in designing strategies for managing risks etc. However, consideration should also be given to other developments or exploration in this field which could be part of the future works.

References:

Chengying, et al. "Explain systemic risk of commodity futures market by dynamic network." *International Review of Financial Analysis* 88 (2023): 102658

Primiceri, G. 2005. Time-varying structural vector autoregressions and monetary policy. Review of Econometric Studies, 72:821-852.